TEMPEST20240624_H2S

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#SET UP

```r
#Set working directory
# knitr::opts_knit$set(root.dir = "C:/Users/ashmo/OneDrive - Arizona State University/Data/TEMPEST/Pore

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(dplyr)
library(ggplot2)
library(data.table)
```

```
##
## Attaching package: 'data.table'
##
## The following objects are masked from 'package:lubridate':
##
##     hour, isoweek, mday, minute, month, quarter, second, wday, week,
##     yday, year
##
## The following objects are masked from 'package:dplyr':
##
##     between, first, last
##
## The following object is masked from 'package:purrr':
##
##     transpose
```

```r
library(stringr)
library(plater)
```

```r
#Function to calculate slope, intercept, and R-squared and return in data frame
#Fe 2
cal_curve<-function(Conc_uM,ABS){
    model<-lm(ABS~Conc_uM)
    slope<-coef(model)[[2]]
    intercept<-coef(model)[[1]]
     R2<-summary(model)$adj.r.squared
  return(data.frame(slope=slope,intercept=intercept,R2=R2))
}
```

#IMPORT DATA

Files required: CSV for each plate run. Each plate should include a set of standards and QAQC checks (duplicates, spikes, interference)

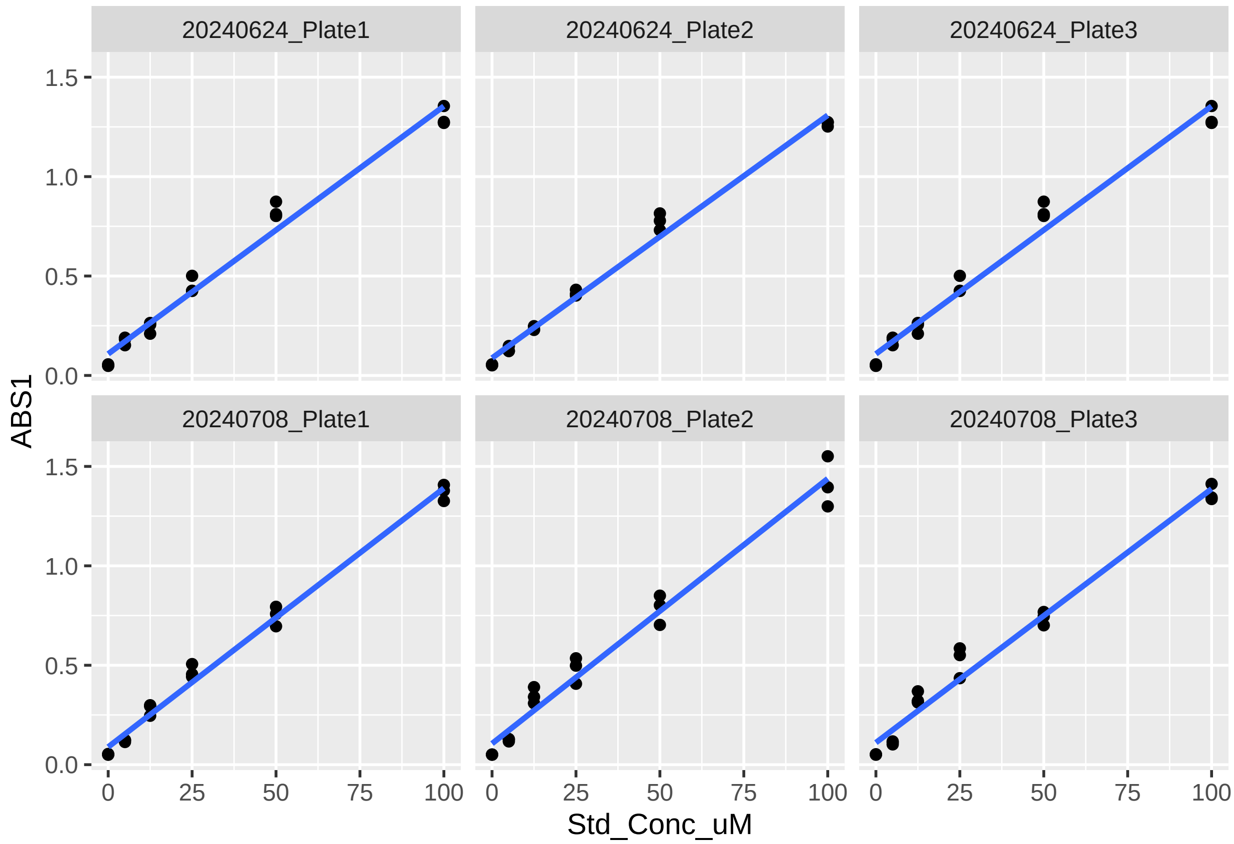
```r
# setwd("C:/Users/ashmo/OneDrive - Arizona State University/Data/TEMPEST/Porewater/02_CleanMicroplateDa

plates<-read_plates(
  files=c("Tidy Data/TEMPEST20240624_H2S_plate1.csv",
          "Tidy Data/TEMPEST20240624_H2S_plate2.csv",
          "Tidy Data/TEMPEST20240624_H2S_plate3.csv",
          "Tidy Data/TEMPEST20240708_H2S_plate1.csv",
          "Tidy Data/TEMPEST20240708_H2S_plate2.csv",
          "Tidy Data/TEMPEST20240708_H2S_plate3.csv"),
  plate_names = c("20240624_Plate1",
                  "20240624_Plate2",
                  "20240624_Plate3",
                  "20240708_Plate1",
                  "20240708_Plate2",
                  "20240708_Plate3"),
  well_ids_column = "Wells",
  sep=",")

head(plates)
```

```
## # A tibble: 6 x 6
##   Plate           Wells  ABS1 IDs                Dilutions Std_Conc_uM
##   <chr>           <chr> <dbl> <chr>                  <int>       <dbl>
## 1 20240624_Plate1 A01   0.048 STD0_0uM                   1           0
## 2 20240624_Plate1 A02   0.056 STD0_0uM                   1           0
## 3 20240624_Plate1 A03   0.05  STD0_0uM                   1           0
## 4 20240624_Plate1 A04   0.051 T0_20240610_FW_C6          1          NA
## 5 20240624_Plate1 A05   0.051 T0_20240610_FW_C6          1          NA
## 6 20240624_Plate1 A06   0.051 T0_20240610_FW_C6          1          NA
```

##PLOT STANDARDS

```r
plates%>%
  filter(str_detect(plates$IDs,"STD"))%>%
  ggplot(aes(Std_Conc_uM,ABS1))+
  facet_wrap(~Plate)+
  geom_point()+
  geom_smooth(method="lm",se=FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
stds<-plates%>%
  filter(str_detect(plates$IDs,"STD"))%>%
  group_by(Plate)%>%
  mutate(cal_curve(Std_Conc_uM,ABS1))%>%
  filter(row_number()==1)%>%
  select(Plate,slope,intercept,R2)

stds
```

```
## # A tibble: 6 x 4
## # Groups:   Plate [6]
##   Plate            slope intercept    R2
##   <chr>            <dbl>     <dbl> <dbl>
## 1 20240624_Plate1 0.0125    0.108  0.979
## 2 20240624_Plate2 0.0122    0.0876 0.989
## 3 20240624_Plate3 0.0125    0.108  0.979
## 4 20240708_Plate1 0.0130    0.0890 0.991
## 5 20240708_Plate2 0.0133    0.106  0.974
## 6 20240708_Plate3 0.0128    0.111  0.975
```

#CALCULATE CONCENTRATIONS

```r
samples<-left_join(plates,stds,by="Plate")%>% #add slope/intercept info
  filter(!str_detect(plates$IDs,"STD|DUP|SPK|ppt"))%>% #filter out standards and checks
  mutate(conc_uM=((ABS1-intercept)/slope)*Dilutions)%>% #calculate concentrations
  mutate(conc_uM=ifelse(conc_uM<=0,0,conc_uM))%>% #make negative values 0
  group_by(IDs)%>%
  summarise(conc_uM=mean(conc_uM))

head(samples)
```

```
## # A tibble: 6 x 2
##   IDs                          conc_uM
##   <chr>                          <dbl>
## 1 20230611_AFTERNOON_SOURCE_SW       0
## 2 20230611_MIDDAY_SOURCE_SW          0
## 3 20230611_MORNING_SOURCE_SW         0
## 4 20240611_AFTERNOON_SOURCE_FW       0
## 5 20240611_HIGH_SOURCE_ESTUARY       0
## 6 20240611_LOW_SOURCE_ESTUARY        0
```

#EXPORT

```r
# setwd("C:/Users/ashmo/OneDrive - Arizona State University/Data/TEMPEST/Porewater/04_SummarizedData")

write.csv(samples, file="Processed Data/TEMPEST2024_H2S.csv")
```